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The relationship of methylation and mRNA expression profile of *AUTS2* with suicidal ideation in adolescents with bipolar depression

Xiaoqin Shen¹, Chengji Wang², Alimire Alimujiang², Cheng Zhang¹, Shaohong Zou^{1*} and Lu Liu^{3,4*}

Abstract

Objective This study aims to explore the correlation between the methylation levels of the *AUTS2* gene and suicidal ideation in adolescents with bipolar depression, and investigate the mediating effect of *AUTS2* mRNA expression levels.

Methods A total of 73 adolescents diagnosed with bipolar depression were recruited for the evaluation of suicidal ideation using the Suicidal Ideation Attributes Scale (SIOSS) and were categorized into two groups (with or without suicidal ideation). The methylation and mRNA expression levels of the *AUTS2* gene were detected through pyrophosphate sequencing and quantitative reverse transcription polymerase chain reaction (qRT-PCR) respectively, and were analyzed for the association with suicidal ideation by group comparison, correlation analyses and mediation analyses.

Results Among 17 methylation sites identified, the methylation levels at the CpG14.15.16 loci were elevated in the group with suicidal ideation compared to those without such thoughts. The mRNA expression level of *AUTS2* was also found be higher in the group with suicidal ideation. Quantitative analyses indicated significant correlation between the methylation level and suicidal ideation, between the methylation level of CpG14.15.16 loci and *AUTS2* expression, and between its expression level and suicidal ideation. Further mediation analyses indicated that the *AUTS2* mRNA expression levels mediated the relationship between the methylation levels at CpG14.15.16 and suicidal ideation, accounting for 30% of the observed effect.

Conclusion The elevated methylation level of CpG14.15.16 of the *AUTS2* gene might be involved in the pathophysiology of suicide ideation in adolescents with bipolar depression, which might be mediated by the mRNA expression of *AUTS2*.

Keywords Adolescent bipolar depression, Suicidal ideation, *AUTS2* gene, DNA methylation, mRNA expression levels

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Introduction

Adolescent bipolar disorder is marked by recurrent episodes of mania and depression, accompanied by an elevated risk of suicidal behavior [1], which the weighted average prevalence is estimated at 3.9% [2]. More than half of them initially experience a depressive episode, who may face higher rates of illness, death and suicide risk than those with bipolar mania, which contributes to long-term functional impairment, psychological disorders, and reduced work efficiency [3]. For adolescents, those experiencing bipolar depression exhibit significantly greater levels of suicidal tendencies, anhedonia, and feelings of hopelessness when compared to their counterparts with unipolar depression. Notably, 60% of adolescents suffering from bipolar depression engage in suicidal behaviors, in contrast to 30% of those diagnosed with unipolar depression. Furthermore, these individuals frequently present with higher incidences of comorbid conditions, such as disruptive behavior disorders, anxiety disorders, and substance use disorders [4]. A large cross-sectional study revealed alarming rates of 12.7% for suicidal ideation and 35.6% for suicide attempts among adolescents with bipolar depression [5]. Despite the evident correlation between adolescent bipolar depression and suicidal ideation, the underlying pathophysiological mechanisms that contribute to the heightened suicide risk remain largely unexplored.

Epigenetic mechanisms, particularly DNA methylation, has been suggested to be involved in the pathophysiology of both bipolar depression, and suicide behavior or suicide ideation [6, 7]. For bipolar depression, one study found that women with bipolar depression show elevated global DNA methylation (5-mC, LINE-1, Alu) compared to controls, with correlations to symptom severity [8]. Besides, no significant difference in methylation levels was found for patients with bipolar depression compared to healthy controls in two recent studies [9, 10]. However, most of the existing studies have focused on adults, with limited exploration of DNA methylation in adolescents. One notable study investigated the epigenetic characteristics of at-risk adolescents for familial bipolar disorder, revealing significant differences between groups in DNA methylation, and polygenic risk scores. This finding implies that environmental factors might interact with genetic susceptibility through epigenetic mechanisms, affecting the onset of psychiatric disorders in adolescents, especially bipolar disorder [11]. In a study [12], they analyzed genome-wide expression and methylation levels of peripheral blood mononuclear cells in children and adolescents from three matched groups: those diagnosed with bipolar disorder, unaffected offspring of bipolar parents (high risk), and low-risk controls. This analysis identified 43 genes that may distinguish patients and at-risk youth from controls, highlighting the potential role of

DNA methylation in the risk of bipolar disorder among youth. Notably, bipolar depression was not explored in those studies for adolescents.

Altered DNA methylation has also been suggested to be linked with suicidal behavior and ideation, which often co-occurred in adolescents with bipolar depression. One study for military veterans found a significant correlation between the hypomethylation of *SKA2* and current suicidal ideation and behavior [13]. Another study examined the link between genome-wide methylation and the risk of current suicidal ideation. It discovered that higher methylation levels of CpG cg06121808 in *SLC20A1* gene on chromosome 2 were associated with current suicidal ideation [14]. Furthermore, a study investigating the methylation patterns related to suicidal ideation that those with such thoughts had significantly higher levels of *MSK1* methylation than those without. Additionally, the methylation rate of *MSK1* was a significant predictor of suicidal ideation among adolescents with major depression [15]. Currently, there is insufficient evidence regarding how methylation affects suicidal thoughts in adolescents with bipolar depression, which still need further investigation. We have conducted preliminary exploration to elucidate this important question.

In the initial phase, we employed Illumina HD 850 K chip technology to conduct a high-throughput screening of genome-wide methylation for 10 adolescents diagnosed with bipolar depression, including 6 patients with suicidal ideation, and 4 patients without. There were significant differences in the methylation of *ABI3BP*, *HLA-DQB1*, *HLA-DRB1*, *AUTS2*, *SP3*, *NINJ2*, *DPYSL2*, and other genes between the suicidal and non-suicidal ideation groups [16]. Among these genes, *AUTS2* has been considered to be an important one for psychiatric disorders, including bipolar disorder and suicide. Briefly, prior research has suggested a potential link between the *AUTS2* gene and bipolar disorder [17]. Meanwhile, the *AUTS2* gene was also associated with suicide. The genetic polymorphism of rs6943555/*AUTS2* was found to be associated with suicide risk [18]. In addition, a genetic study focused on family lineages with a high risk of suicide identified variations in the *AUTS2* gene [19]. Based on these existing evidences, it was worth exploring whether the methylation of *AUTS2* participated in the co-occurrence of bipolar depression and suicide ideation in adolescents. In addition, the potential mechanism of the altered methylation level needs more explanation, such as the effect of methylation on gene expression.

DNA methylation represents an epigenetic modification that modulates gene expression through the addition of methyl groups to the C5 position of cytosine, particularly within CpG dinucleotides [20]. A recent investigation demonstrated that hypermethylation at two distinct CpG sites within the *TrkB* promoter region in the frontal

cortex (specifically in Brodmann areas 8 and 9) led to a decrease in the expression of TrkB T1 subtype mRNA, which was associated with suicidal behavior [21].

Drawing from prior molecular genetic investigations, this study specifically targeted adolescent patients diagnosed with bipolar depression, focused on the co-occurrence of suicide ideation and investigated the potential alteration of DNA methylation of *AUTS2* gene and the potential mediation role of its mRNA expression level. The findings would deepen the comprehension of the pathological mechanisms underlying suicidal ideation in adolescents suffering from bipolar depression.

Methods and materials

Subjects

A total of 73 adolescents diagnosed with bipolar depression were recruited from the Clinical Psychology Department of the People's Hospital of Xinjiang Uygur Autonomous Region. The criteria for inclusion in the study were as follows: (1) aged between 12 and 18 years; (2) meeting the diagnostic criteria for bipolar disorder as specified in the DSM-5 disorder (currently experiencing a depressive episode that has lasted for 2 weeks or more); (3) the score of the Young Mania Rating Scale (YMRS) [22] ≤ 7 and the score of Hamilton Depression Rating Scale (HAMD-24) [23] ≥ 20 ; (4) had no additional mental health comorbidities; (5) clinically diagnosed by two experienced psychiatric physicians (deputy chief or chief physicians); (6) All our subjects are Han Chinese. Exclusion criteria were: (1) comorbid with other Axis I or Axis II mental health disorders; (2) central nervous system disorders, endocrine conditions that influence mood, or diseases affecting the immune system; (3) a documented history of substance abuse or dependence; (4) pervasive developmental disorders; (5) intelligence quotient (IQ) ≤ 80 .

The YMRS and other clinical scales were evaluated by psychiatrists who received standardized training in clinical. They also received two hours of training on the objectives and procedures of the investigation.

This study received approval from the Ethics Committee of the People's Hospital of Xinjiang Uygur Autonomous Region (Ethics batch number: KY2019120606). All participants, along with their legal guardians, provided informed consent to participate in the study by signing a consent form.

Sample size

The sample size was calculated using PASS15. The calculations were based on an α level of 0.05, power of 0.9. The minimum required sample size was determined to be 18 cases in each group. A total of 73 adolescents were enrolled in this study, which satisfied the minimum sample size required for this study.

Assessment of suicidal ideation

The Self-Rating Idea of Suicide Scale (SIOSS) comprises four dimensions: despair, optimism, sleep, and disguise. This scale consists of 26 items, each of which is answered with either "yes" or "no." For items numbered 2, 3, 4, 8, 11, 12, 14, 16, 17, 18, 19, 20, 22, 23, 24, and 26, a "Yes" response is assigned 1 point, while a "No" response receives 0 points. Conversely, for items 1, 5, 6, 7, 9, 10, 13, 15, 21, and 25, a "No" response earns 1 point, and a "Yes" response earns 0 points. A total score of 12 or higher, with a masking factor score of less than 4, indicates the presence of suicidal ideation; a total score below 12 signifies the absence of such ideation [24].

The collection of whole blood samples

5 ml of elbow venous blood was collected from participants, placed in an EDTA anticoagulation tube, labelled. Among these, 2 ml of whole blood was used for DNA extraction, PBMC was isolated from the remaining 3 ml, and RNA was extracted from peripheral blood PBMC using the Trizol method, then both were stored in a refrigerator at -80°C .

Determination of *AUTS2* gene methylation levels

The Methprimer software [25] was utilized to forecast potential methylation islands within the body region of the *AUTS2* gene and to design the relevant methylation primers. The primer sequences are as follows: 5' terminal primer sequence: aggaagagag GGGGGTTTTTTATTGT TTTTTTTGT; 3' end primer sequences: cagtaatcgact-cactataggagaaggct AAAACCCACACTTAACTCC T. Following the extraction of DNA, bisulfite treatment was performed prior to PCR amplification. The resulting PCR products were validated through electrophoresis and subsequently sent to a biological company for pyrophosphate sequencing of the target fragment, identification of methylated CpG sites and the calculation of the methylation rate, thereby allowing for the assessment of the methylation level of the *AUTS2* gene.

Determination of *AUTS2* Gene mRNA expression levels

After the RNA was extracted and qualified, reverse transcription was conducted to synthesize complementary DNA (cDNA). Primer5 software was employed to design the primers, which are detailed as follows: *AUTS2*-F: GCGCTGGAGAAAGATGTAGC, *AUTS2*-R: CAAGGTGCTGGCAGAGATTG. Quantitative fluorescence experiments were carried out, and quantitative reverse transcription PCR (qRT-PCR) was utilized to measure the mRNA expression levels of the *AUTS2* gene with β -actin as reference gene, then the target gene expression level was calculated using the $2^{-\Delta\Delta\text{CT}}$ method.

Statistical analysis

The statistical evaluation was performed utilizing SPSS version 26.0. For datasets exhibiting a normal distribution and equal variances, results were expressed as mean $\pm$ standard deviation (Mean $\pm$ SD), independent-samples t test and covariance analysis were employed for comparisons between groups. In cases where the data did not conform to a normal distribution, we reported the median along with the interquartile range [M (P25, P75)], and the Mann–Whitney U test was applied for inter-group comparisons. Categorical data were presented as frequencies and percentages (%), with χ^2 tests utilized for group comparisons.

To investigate the relationship between the methylation levels and mRNA expression levels of the *AUTS2* gene, we employed partial correlation analysis. Furthermore, we assessed the correlation with SIOSS scores in adolescent patients diagnosed with bipolar depression. The mediating effect was examined using Model 4 from the SPSS macro Process, specifically analyzing the mediating role of mRNA expression levels between gene methylation levels and suicidal ideation, in accordance with the Bootstrap method as outlined by Hayes [26]. A two-tailed significance level of $P < 0.05$ was established as the threshold for statistical significance.

Results

Comparison of general data between the group with and without suicidal ideation

Among the 73 adolescents diagnosed with bipolar depression, a total of 49 individuals (67.12%) reported experiencing suicidal ideation, whereas 24 individuals (32.88%) did not exhibit such thoughts. The two groups were found to be comparable, with no significant differences observed in terms of gender, age, age of onset, or course of disease ($P > 0.05$) (Table 1).

Comparison of methylation and mRNA expression levels of *AUTS2* gene between suicidal ideation group and non-suicidal ideation group

A total of 17 methylation sites was identified for *AUTS2* gene using pyrophosphate sequencing, including CpG1, CpG6.7, CpG8, CpG9, CpG10, CpG14.15.16, CpG24.25, CpG30.31, CpG32.33.34, CpG35.36.37, CpG42.43, CpG45.46, CpG48.49, CpG50.51.52, CpG53, CpG54, and CpG56.

Through covariance analysis with sex, age, age of onset and course of disease controlled, it was found that the methylation levels at the CpG1 ($P = 0.031$), CpG10 ($P = 0.033$), and CpG14.15.16 ($P = 0.009$) sites were markedly elevated in the group exhibiting suicidal ideation (Table 2). However, these difference could not survive false discovery rate (FDR) correction for multiple testing using ($P_{\text{FDR}} > 0.05$).

For mRNA expression, the expression level of *AUTS2* in patients with suicidal ideation was higher than that without suicidal ideation ($P = 0.005$) (Table 2).

Correlation of *AUTS2* gene methylation and mRNA expression levels with suicidal ideation

Through partial correlation analysis with sex, age, age of onset and course of disease controlled, the correlation analysis demonstrated a positive association between the methylation levels at the CpG14.15.16 and the mRNA expression levels ($r = 0.256$, $P = 0.034$), as well as with the SIOSS score ($r = 0.243$, $P = 0.045$). Furthermore, a positive correlation was observed between the mRNA expression levels and the SIOSS score ($r = 0.335$, $P = 0.005$) (Fig. 1).

Mediation analyses for *AUTS2* gene methylation, mRNA expression levels, and suicidal ideation

Using PROCESS, the mediation effect of mRNA expression level (M) on the relationship between methylation level of CpG14.15.16 (X) and Suicidal ideation (Y) was evaluated. The result showed that the direct path from

Table 1 Comparison of the general data between the two groups

	with Suicidal ideation (N = 49)	without Suicidal ideation (N = 24)	t/ χ^2 /Z	P-value
^c Gender (n, %)			0.183	0.669
male	12 (24.49)	7 (29.17)		
female	37 (75.51)	17 (70.83)		
^a Age (years, mean $\pm$ SD)	15.51 $\pm$ 2.04	16.04 $\pm$ 2.18	1.022	0.310
^a Age of onset (years, mean $\pm$ SD)	13.86 $\pm$ 2.30	14.29 $\pm$ 2.79	0.706	0.482
^b Course of disease [Month, M (P25, P75)]	12.00 (3.00, 24.00)	18.00 (7.25, 36.00)	-1.037	0.300

^aIndependent-samples t test

^bMann–Whitney U test

^c χ^2 tests

Table 2 Comparison of methylation and mRNA expression levels of *AUTS2* gene between two groups (mean $\pm$ SD)

	with Suicidal ideation (N=49)	without Suicidal ideation (N=24)	F	P-value
Methylation level (CpG island methylation rate%)				
CpG1	9.258 $\pm$ 1.175	4.723 $\pm$ 1.683	4.847	0.031
CpG6.7	7.776 $\pm$ 1.265	5.874 $\pm$ 1.812	0.736	0.394
CpG8	4.343 $\pm$ 0.696	3.424 $\pm$ 0.997	0.568	0.454
CpG9	7.908 $\pm$ 0.997	5.021 $\pm$ 1.429	2.728	0.103
CpG10	11.130 $\pm$ 1.633	4.901 $\pm$ 2.340	4.732	0.033
CpG14.15.16	6.158 $\pm$ 0.725	2.720 $\pm$ 1.039	7.314	0.009
CpG24.25	10.734 $\pm$ 1.269	6.710 $\pm$ 1.818	3.272	0.075
CpG30.31	11.217 $\pm$ 1.768	5.806 $\pm$ 2.533	3.048	0.085
CpG32.33.34	9.180 $\pm$ 1.248	6.840 $\pm$ 1.788	1.144	0.289
CpG35.36.37	5.069 $\pm$ 0.897	3.401 $\pm$ 1.286	1.123	0.293
CpG42.43	7.160 $\pm$ 1.011	4.841 $\pm$ 1.449	1.710	0.195
CpG45.46.47	9.457 $\pm$ 0.969	8.818 $\pm$ 1.389	0.141	0.708
CpG48.49	7.313 $\pm$ 0.956	4.736 $\pm$ 1.369	2.364	0.129
CpG50.51.52	9.487 $\pm$ 1.088	6.339 $\pm$ 1.559	2.722	0.104
CpG53	7.391 $\pm$ 1.293	3.743 $\pm$ 1.852	2.589	0.112
CpG54	8.467 $\pm$ 2.781	10.422 $\pm$ 3.985	0.161	0.690
CpG56	9.203 $\pm$ 2.771	8.628 $\pm$ 3.970	0.014	0.906
mRNA expression level				
<i>AUTS2</i>	2.949 $\pm$ 0.380	1.035 $\pm$ 0.544	8.257	0.005

methylation level of CpG14.15.16 to suicidal ideation did not reach statistical significance, but the indirect path through mRNA expression level was significant [$B = 0.68$ ($SE = 0.26$), 95% $CI = 0.16 - 1.21$]. In summary, a complete mediation model was detected, showing that mRNA expression level mediated the path between methylation level of CpG14.15.16 and Suicidal ideation (Fig. 2).

Discussion

This research aimed to investigate how *AUTS2* gene methylation relates to suicidal ideation in adolescents with bipolar depression and to analyze the mediating role of *AUTS2* gene mRNA expression in this relationship. The results indicated that the group with suicidal ideation exhibited higher levels of *AUTS2* gene methylation and expression. The *AUTS2* mRNA expression level mediated the relationship between *AUTS2* gene methylation and suicidal ideation.

The findings showed that individuals with suicidal ideation exhibited increased methylation levels at specific sites of the *AUTS2* gene, including CpG1, CpG10, CpG14.15.16. A significant focus of previous research has been the *AUTS2* gene's impact on neurodevelopmental disorders, as well as its association with cognitive

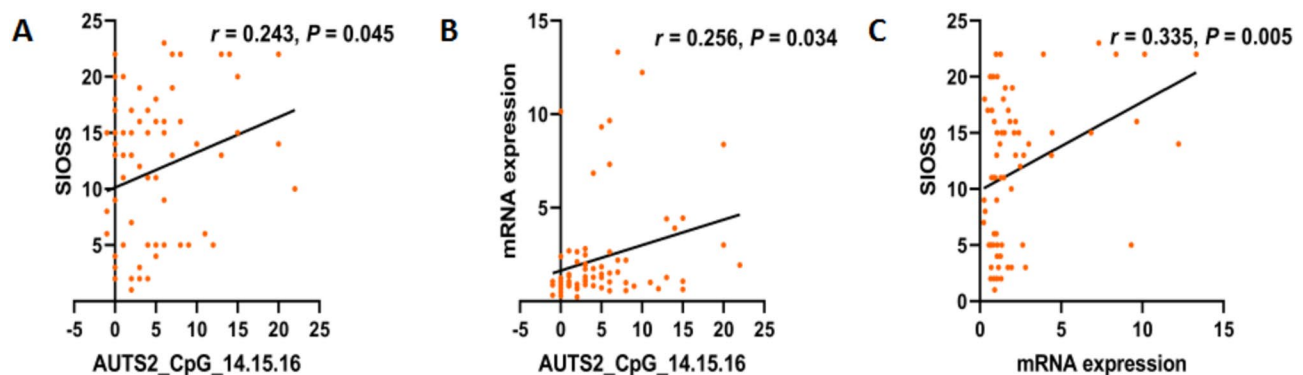


Fig. 1 Correlation analysis of *AUTS2* gene methylation and mRNA expression levels with suicidal ideation. (A) Positive correlation between methylation levels of CpG14.15.16 in the *AUTS2* gene and SIOSS scores. (B) Positive correlation between methylation levels of CpG14.15.16 in *AUTS2* gene and *AUTS2* mRNA expression levels. (C) Positive correlation between *AUTS2* mRNA expression level and SIOSS scores

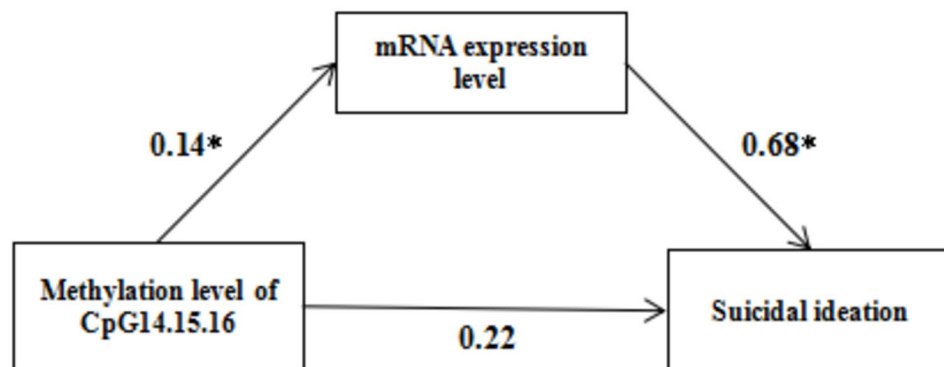


Fig. 2 The mediation effect of *AUTS2* mRNA expression level on the relationship between methylation level of CpG14.15.16 and suicidal ideation

function and behavioral characteristics. A research investigation into the histone H3K4 methylation pathway genes, has revealed a significant association between the *AUTS2* gene and 18 brain features. These traits encompass alcohol and coffee consumption, propensity for risk-taking, smoking patterns, brain volume, cognitive intelligence, reaction times, self-reported educational attainment, levels of happiness, and mental disorders, including bipolar disorder [27]. Previous studies on the methylation of the *AUTS2* gene have primarily focused on the following areas: placental epigenetic inheritance, and psychological treatments for post-traumatic stress disorder. A study found that *AUTS2* methylation patterns in the placenta are linked to maternal serum PCBS levels and child neurodevelopment, indicating that *AUTS2* methylation could be a marker of altered placental function or increased Autism spectrum disorder (ASD) risk following maternal PCBS exposure [28]. In addition, a study on psychotherapy for post-traumatic stress disorder found that increased methylation of the *AUTS2* gene was linked to reduced severity of PTSD symptoms in patients treated with forensic Offender Rehabilitation Narrative Exposure Therapy (FORNET) [29]. Alarming, there is a lack of research on how *AUTS2* gene methylation may influence suicidal ideation in adolescents with bipolar depression. Our study extends this knowledge, finding that adolescents with suicidal ideation have significantly higher methylation levels at specific CpG sites in the *AUTS2* gene (CpG1, CpG10, CpG14,15,16) compared to adolescents without suicidal ideation. This finding suggests that *AUTS2* gene methylation may be a potential biomarker of suicidal ideation in adolescents with bipolar depression.

The findings of this investigation indicate that the mRNA expression levels of the *AUTS2* gene are significantly elevated in individuals experiencing suicidal ideation compared to those who do not exhibit such thoughts. Correlation analysis revealed a positive association between the methylation levels at CpG sites 14,15,16 of the *AUTS2* gene and its corresponding mRNA expression levels. Previous analyses of *AUTS2* gene expression have not linked it to bipolar depression or suicidal ideation in adolescents. However, it has been studied in relation to other conditions, such as heroin dependence and hypertrophic cardiomyopathy, as well as in animal studies. A genetic study found significantly lower levels of *AUTS2* expression in lymphoblast-like cell lines (LCL) from individuals dependent on heroin compared to those in the control group, and further genetic association analysis revealed that the *AUTS2* mRNA level in LCL from rs6943555 AA carriers ($n=19$) was significantly lower than in TT carriers ($n=97$, $p=0.002$) and AT carriers ($n=91$, $p=0.005$). This indicates that decreased *AUTS2* gene expression may increase susceptibility to

heroin dependence [30]. Animal studies indicated that the mRNA of the *AUTS2* gene was highly expressed in the developing cerebral cortex, cerebellum, and several other brain regions in mice. On embryonic day (E) 12, the mRNA of the *AUTS2* gene was found in cortical prelates. From E16 to 21 days after birth, it was most abundantly expressed in the frontal cortex, hippocampus, and cerebellum. High levels were also observed in the developing dorsal thalamus, olfactory bulbs, hypothalamus, and substantia nigra [31]. In addition, a methylation study of hypertrophic cardiomyopathy (HCM) demonstrated a significant positive correlation ($r=0.70$, $p=0.0093$) between the methylation of the *AUTS2* gene (cg20713355) and its expression levels [32]. Research has indicated that a majority of CpG sites situated in proximity to the transcription start site (TSS) of promoters possess the ability to hinder the binding of transcription factors or diminish their affinity, particularly at elevated levels of methylation. This suppression can result in a decrease in gene transcription and a subsequent reduction in gene expression levels. On the contrary, when the methylation level of CPG in the gene body which far from TSS increases, the gene expression level increases [33]. Consequently, the impact of DNA methylation on gene expression-whether it serves to inhibit or activate-primarily hinges on the specific genomic location as well as the nature of the external stimulus [34]. This indicates that promoter methylation is inversely correlated with gene expression, while methylation within the gene body exhibits a positive correlation with expression levels [35]. According to the high-throughput sequencing results of the pre-test in this study [16], the cg09994954 locus of *AUTS2* gene that is different between the suicidal ideation group and the non-suicidal ideation group is located in the gene body region, so when the methylation level of *AUTS2* gene increases, the gene expression level also increases. This phenomenon may be attributed to the role of gene body methylation in diminishing the likelihood of allele-specific expression via cis-regulatory mechanisms. Additionally, gene body methylation contributes to the stabilization of transcription by inhibiting the incorporation of the histone variant H2A.Z, which in turn enhances and maintains elevated expression levels [36].

This investigation further examined the relationship between the methylation of the *AUTS2* gene and suicidal ideation by analyzing continuous variables through inter-group comparisons. The findings demonstrated a positive association between the methylation levels at CpG sites 14, 15, 16 of the *AUTS2* gene and the SIOSS score, thereby partially validating the results obtained from inter-group comparisons. Mediation analysis indicated that the mRNA expression levels of the *AUTS2* gene served as a mediator in the relationship between

the methylation levels at CpG sites 14, 15, 16 and suicidal ideation, explaining 30% of the observed effect. This finding implies that mRNA expression may have a mediating function in the interplay between gene methylation and suicidal ideation. Previous studies have found that neurodevelopment may be a physiological process affected by suicidal ideation, that is, molecular pathways related to neuron migration and proliferation are associated with suicidal ideation [37]. The *AUTS2* gene is fundamentally associated with a range of neuropsychological disorders and is integral to numerous neurodevelopmental processes. Within the nucleus, *AUTS2* functions as an essential transcriptional regulator involved in neurodevelopment. In the cytoplasmic compartment, it plays a pivotal role in cerebral corticogenesis, encompassing neuronal migration and neurocytogenesis, through the modulation of cytoskeletal dynamics. Postnatally, *AUTS2* is responsible for modulating the number of excitatory synapses, thereby maintaining the equilibrium between excitatory and inhibitory signals within neural circuits [38, 39]. Collectively, these results lend support to the hypothesis that epigenetic modifications of the *AUTS2* gene are crucial in understanding the pathophysiological mechanisms underlying suicidal ideation in adolescents suffering from bipolar depression.

The results of this study confirmed that *AUTS2* gene methylation can regulate the mRNA expression level, and then affect the suicidal ideation of adolescents with bipolar depression. The possible biological mechanism or pathway is as follows: Firstly, research highlights that *AUTS2* exhibits high expression levels in the frontal, parietal, and temporal lobes of the fetal brain, as well as in various neuronal subtypes, including glutamatergic, GABAergic, and tyrosine hydroxylase (TH)-positive dopaminergic neurons [40]. Methylation of the *AUTS2* gene has the potential to induce an imbalance in neurotransmitter levels, particularly those critical for the regulation of emotions, such as dopamine and glutamate. This alteration may influence synaptic efficacy and connectivity, leading to heightened sensitivity to stress in individuals, which could, in turn, precipitate suicidal ideation. Furthermore, the expression of *AUTS2* is notably prominent in various types of brain cells, with a particular emphasis on astrocytes. These glial cells playing a crucial role in the development and functionality of synapses and neural circuits during early brain maturation [41]. FMRI for MDD and other mood disorders also demonstrated disruption of neural circuits [42, 43], suggesting that the absence of *AUTS2* in astrocytes may contribute to the pathogenesis of brain disorders by impairing synaptic formation during developmental stages. The methylation of the *AUTS2* gene may interfere with the development and functionality of neural circuits that are essential for the transmission and processing of

information. Such disruptions can adversely affect emotional regulation and cognitive capabilities, thereby elevating the risk of suicidal ideation. Additionally, animal experiments demonstrated that *AUTS2* is located in both the nucleus and the dendritic spine. In dendritic spines, *AUTS2* interacts with the AKT E3 ligase TTC3, regulating synaptic AKT/mTORC1 signaling and AKT-related ubiquitination. Defects in *AUTS2* can disrupt synaptic connectivity and contribute to excitatory synaptic dysfunction. Furthermore, research has indicated that the aberrant regulation of AKT-mTOR signaling within the hippocampus of mice plays a significant role in the pathophysiology of depression [44]. The *AUTS2* gene may be implicated in the pathophysiological mechanisms underlying bipolar depression and associated suicidal ideation through its regulation of the AKT/mTOR signaling pathway. The outcomes of this investigation offer critical insights that could inform future studies on the implications of DNA methylation and mRNA expression levels concerning suicidal thoughts in adolescents diagnosed with bipolar depression.

Several potential limitations of the present study should be taken into consideration. First, a key limitation of this research is the small sample size of 73 adolescents with bipolar depression, which may limit the generalizability of our findings. Second, the cross-sectional design of the study also limits our ability to establish causal relationships among *AUTS2* gene methylation levels, mRNA expression, and suicidal ideation. Future research should focus on larger multicentre and longitudinal studies. Furthermore, the reliance on self-reported measures of suicidal ideation introduces the potential for bias, which could compromise the accuracy of the results. Our findings may be population-specific and need for future research to validate these results in diverse populations. These studies will help confirm the findings and improve our understanding of the mechanisms behind suicidal ideation in adolescents with bipolar depression.

Conclusion

Our findings show a positive correlation between *AUTS2* gene methylation at CpG sites 14,15,16 and suicidal ideation. Additionally, mRNA expression levels of the *AUTS2* gene seem to mediate this relationship, indicating a possible pathway linking gene methylation to suicidal ideation. These findings highlight the crucial role of epigenetic modifications in the pathophysiology of suicidal ideation and suggest that the *AUTS2* gene could serve as a biomarker for identifying individuals at higher risk.

Acknowledgements

The authors would like to acknowledge all research assistants and investigators for their dedicated assistance in the data collection. We would especially like to thank our participants in the study for their time and willingness to participate.

Author contributions

Shaohong Zou conceived and designed the study. Lu Liu directed and revised the article. Xiaoqin Shen wrote and revised the article. Xiaoqin Shen, Chengji Wang and Alimire Alimujiang collected the data. Cheng Zhang performed the clinical assessment. All authors contributed to and have approved the final manuscript.

Funding

This work was supported by the Natural Science Foundation of Xinjiang Uygur Autonomous Region (No: 2022D01C609), the Tianshan Innovation Team Plan (No: 2022D14011).

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the Institutional Review Board (IRB) of People's Hospital of Xinjiang Uygur Autonomous Region. All participants, along with their legal guardians, provided informed consent to participate in the study by signing a consent form. All experiments were performed in accordance with relevant guidelines and regulations. All experiments were performed in accordance with the declaration of Helsinki.

Consent for publication

Not Applicable.

Competing interests

The authors declare no competing interests.

Received: 9 January 2025 / Accepted: 30 April 2025

Published online: 26 May 2025

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